

Business Plan

As an example or template of a business plan, I took Robinson Crusoe by Defoe, replacing only the archaic Good and Evil with Structured and Unstructured areas.

Unstructured area

The founder is located in St. Petersburg, Russia.

No one will do business with me, as I'm currently on Sanctions Island.

My financial situation is such that I can't afford major expenses.

This situation has been building for years and decades, during which I've never been able to raise funds for my project.

I don't have a patent. I once applied for a provisional patent with the USPTO, but the deadlines have long since passed. And although I had this application in hand for a year, I couldn't raise funds.

My idea occurred to me alone. But maybe that's simply because no one wants it?

My idea may have been brilliant, but everyone who came across it probably asked themselves, "What am I going to do with this?"

I don't have a team, because it's hard to imagine assembling a team under these circumstances.

I don't have a company or a bank account.

So what am I supposed to do with it?

I need to open a company like all startups do, and also a bank account, but I don't have the funds.

It's a vicious cycle.

Structured area

But my product is already in demand in the market, and even those who don't know about it already feel the need for something that will help them better master AR devices.

Many people already know me, and I'm the first search result for "mouse design develop" on Google.

Why is my mouse important specifically for AR devices?

Because now mastering touch typing is as easy as was learning use with computer mouse. My mouse has all the functionality of a keyboard. But since the letters are located on the edges and text is entered by tilting the tip of the pyramid toward the desired letter, you see on the screen, along with the text editor, how you touch the tip of the pyramid before you start typing. This way, you don't have to look at the device itself. It's just like using a mouse without looking at your hand holding the mouse.

What market niche can I fill?

The entire one that augmented reality devices were hoping to fill, but couldn't. And this market niche is no smaller than that of laptops and phones. With a keyboard, AR glasses become a fully-fledged computer.

Of course, I don't have the resources to organize production on my own to produce such a quantity, but I invite anyone interested to participate under the Creative Commons open license. Therefore, I'm an Open Project. My project is located on GitHub, and anyone can create their own development branch. Like any similar project, I require support and sponsorship.

I'm also planning to launch my own commercial project now.

The sanctions don't affect me personally. I have a valid passport and there are no obstacles to traveling to a country where I can set up an LLC or C-Corp for my startup. I need \$6,500-8,100 for this. I can get a loan somewhere and get it done quickly.

This is the entire amount, including plane tickets and accommodation, as well as those needed to open a company and a bank account.

I just understand that this wouldn't be very good for my potential investor, as it would include certain obligations to a third party in our agreement. It would be better to take this amount from sponsorship donations. Such sponsorship donations may be the amount from the investments themselves.

So, a company can be created within a week.

But where will the company be physically located?

So far, I have an invitation from SOSV. For now, I plan to rent desks there, which will cost about \$40,000 per year. An additional \$20,000 will go toward purchasing equipment; more details are included in the financial plan. Offers from other venues are also likely to be considered.

I also plan to hire a couple of employees.

Hiring employees is a complex matter, and it can slow down the entire process. But there's no need to rush it. I already have several candidates. I'm a regular member of a specialized developer portal, specifically the section where they offer jobs. It's quite possible to get a specialist there for \$60,000 in about a month or two. One college is also offering interns, but it's still under negotiation. I've suggested that if anyone shows up, we could create their own development branches on GitHub. The college promises to handle all the logistics, but their salaries should be factored into the expenses, just in case.

The first thing I need to do is create a pilot batch or signal batch of devices. I believe the design of my device is so advanced that I can avoid the need for a series of prototypes. There will also be prototypes, but in the software space. This stage is still fairly early.

Creating a signal batch involves producing several devices, somewhere around 10 units. This also provides a safety net in case the only unit is damaged or lost. I estimate the price of this batch will be around \$50,000, which is detailed in the financial plan and its appendix, where I will try to include all the prices for parts that I have managed to obtain or that seem credible to me.

Copies of the signal batch can be distributed to interested parties and used for demonstration purposes.

I intend to simultaneously improve and develop open source development. These will be the usual activities associated with creating web pages of the required quality. The project will still be divided into open source and commercial parts, but a domain, server, and everything else typically used in such projects are required. The text editor also needs to be developed separately. Perhaps we can also merge with other developers in this area. My text editor differs from others in that it receives two types of data. This is the standard map of codes for all characters, but it also needs its own set of codes for touching the tips of the pyramids before character entry begins. This is what is displayed on the screen before the decision is made to type a particular character.

This is the basis of my Touch Typing, why learning this method will be as easy as using a mouse. But these codes don't yet exist. Therefore, in principle, anyone can create their own, which will inevitably lead to chaos and incompatibility.

Standardization is essential here from the very beginning. And for this, we also need a good website with a clear interface.

Crowdfunding is the next step.

I understand that I can't compete on price with regular mice and keyboards produced in mass quantities. It's possible that my device might even be more expensive than similar products that appear from time to time on crowdfunding platforms. These are typically custom keyboards and mice. Even those who create devices with highly custom designs and ergonomics do so using standard components. These are usually buttons.

Most of my parts are unique. In total, including fasteners, contacts, and springs, my device contains about 150 pieces. All this can be found in more detail in the appendix to the financial plan.

Crowdfunding involves showing a promotional video with a working physical copy. The price of a promotional film is estimated to start at three thousand dollars.

All that remains is to set the sales price, and I've already said that I doubt I'll be able to compete in price not only with mass sales, but also with custom pieces.

Device manufacturers are typically advised to set a selling price three times the cost. These recommendations are typically given to medical equipment manufacturers. These are very small quantities, and the buyer obviously has the ability to pay a high price. My case is the opposite: for a mass consumer. The Chinese also produce in large quantities, but their quantities are huge. I can't immediately budget for millions of units like the Chinese.

What quantity, then, can I budget for? I'll probably plan for a total production run of 100,000 units. The margin I plan to budget for here could be good if it reaches somewhere in the middle, around 10%.

Unstructured area - What if they buy less?

But what if they buy less than a hundred thousand units? That will be a lot less.

Structured area

Many things point to continued, sustained interest, and that the device, one way or another, is something we've long expected on the market. If, however, initial sales are lower than planned, that won't be a red flag. The situation may be fixable in any case. We simply need to understand why this is happening and address the issue.

I plan to gradually enter the market by consistently exploring various crowdfunding platforms.

The usual mistake is that crowdfunders try to sell as much as they're can selling. They don't realize they don't have a warehouse full of finished products.

I don't think stopping early is a bad idea. This will give you the opportunity to assess your capabilities at this stage.

I see a perfectly achievable goal of selling devices at \$200 (199). Of this, 10% would go to the platform itself and another 10% would be margin. This would easily bring the cost price of the device itself to \$160 with a planned production run of 100,000 units.

However, the first trial run would realistically be unprofitable. This means that by releasing 10000 devices at \$200 (199), we would likely continue to burn through the remaining investment funds.

Unstructured area - What if the cost price skyrockets?

What if the cost price skyrockets to \$500, as we see quite often in crowdfunding campaigns, when a \$600 device appears, with a less-than-perfect alternative available in China for less than \$100?

Structured area

I participated in a Chinese design competition. When I made it to the next round, they asked me to send them a physical model. Of course, those were the conditions, but it could also indicate that they're not yet capable of creating my device themselves.

I'm also not opposed to another round of funding, although I don't think it's advisable at this point. If the valuation is favorable, another round is certainly possible.

However, I don't want to raise the selling price, as I envision my buyer as an augmented reality developer. These people, having received the device at a relatively affordable price, will contribute to the project themselves.

Furthermore, it's important to understand that technology is constantly improving and will continue to do so, and at the same time, costs will decrease.

It's best to plan your first crowdfunding campaign on a platform with a proven audience. This is someone interested in new products and looking for ways to use them. I estimate this audience to be around 40,000 -10,000 buyers, and this is the best target for the fundraising. This will generate about \$1.9M in total sales revenue.

For this, we also need to create a promotional video targeting this audience. I estimate the cost here to be around \$3,000 to \$4,000.

Another portion of the funds that will need to be burned will be rent for storage space and the salaries of the employees working there. The assembly of the first batch of 10,000 items can be carried out for now using purely manual assembly.

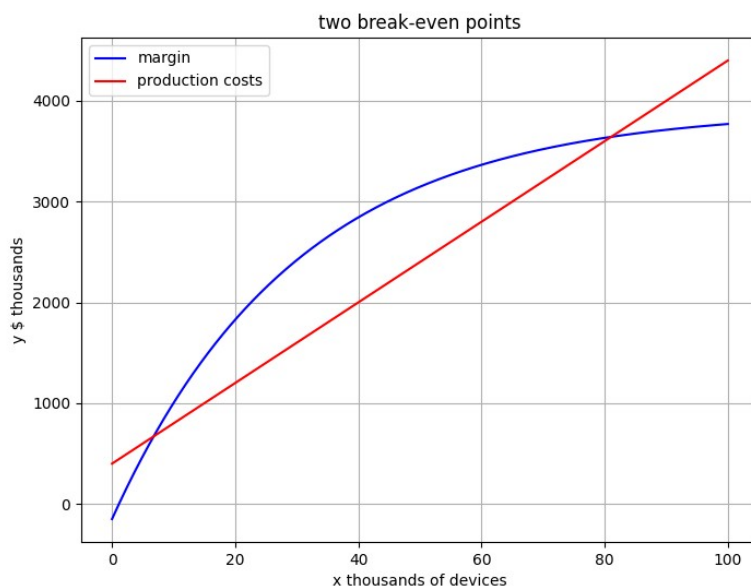
Once the planned \$1,9 or more has been raised, the next step will be ordering components and assembly. Of course, this will also be a complex process, which I will discuss in more detail below.

Unstructured area — margin is an exponent

It turns out that first a first trial stage is planned, after which, after some time, the next fundraising is organized in order to reach the planned goal of 100,000?

Obviously, this second fundraising will be an attempt to recapture previously pent-up demand. Or will it be a way to recapture those who missed the first one? But how successful will this be? It's quite possible that those who didn't manage to buy on the first platform won't return, as their interest will have faded.

Moreover, this second crowdfunding campaign already envisages reaching a break-even point. But will this be achievable? The number of devices sold in advance is growing, which means growth and expansion of production itself are also required. But this growth will likely be extensive, requiring only new capital, hiring, and expanding production and logistics. This is a linear relationship, whereas margin growth is exponential. The line can intersect the exponential at two points. Thus, the first break-even point will signal a transition to profitability, but the next point will mean the process has become unprofitable again.



If first break-even point is planned for when total sales more than 10,000 then when it approaches 80,000 production may again slip into the red. Even with expansion, the original assembly line will no longer be able to handle the larger volume. What will happen next? The board of directors may not be able to handle the assigned tasks. The sales network will fail to develop into an independent entity. The situation may gradually spiral out of control. Attempts to raise further funding rounds to address the shortfall will result in the loss of shares and control.

At the same time, competitors, having assessed the situation, will begin their own production. Their advantage will be felt immediately, as they have long-established production and distribution channels.

Competitors will be well-positioned to take advantage of the situation if I'm late in implementing something.

Another destructive factor can be internal conflict within a company. I've worked in manufacturing my entire life, but I've never held a management position. I can organize my own workplace, but it's probably not so simple with people.

I'll certainly invite staff to the board of directors through LinkedIn profiles with the best references, but how the relationship develops depends on too many subjective factors. There are numerous examples of founders having conflicts with the board of directors.

Another negative factor can be delays and failure to meet deadlines. This could be due to a lack of consideration during planning or unknown external factors specific to the region where the business is located.

Structured area

Even though I haven't started anything yet, many companies have already started offering me their services. Here's a list I compiled of these offers, along with my personal contacts in the required areas of production.

I also have a constantly expanding network of personal contacts on LinkedIn, including those with manufacturing specialists.

<https://www.linkedin.com/in/dellatanpcba/>

<https://www.linkedin.com/in/dinesh-kumar-elangovan-092796a7/>

<https://www.linkedin.com/in/tracy-l-a60982164/>

<https://www.linkedin.com/in/amit-sudarsan-737ab717/>

<https://www.linkedin.com/in/gdusunen/>

<https://www.linkedin.com/in/sahidsesay/>

<https://www.linkedin.com/in/livingasaadhi/>

Here are companies offering various services in the production of parts and components, such as various electronics or plastic molding.

All these companies found me themselves and what's interesting is that my email address is not public.

<https://www.metalworkmfg.com/>

<http://www.zeyatech.com/en>

<https://www.enuomold.com/>

<https://www.majortooling.com/>

<https://www.oemlaptop.com/>

<http://en.jinruiye.com/>

Of course, I'll hire a supply chain manager; they need to have their own supply chain. Otherwise, there's no point in hiring them. But it's always good to have some alternative connections at your fingertips.

If the supplies are parts and components, then assembly requires equipment and space.

Another key employee to hire is a production manager. This also a specialist with the appropriate channels and connections. However, it's important to understand that this person won't be managing the assembly process on their own. Assembly is all about people.

Assembly—the assembly line is a cursed place.

And people work there with the idea that they won't last long. And the complete opposite is true: in assembly, humans have actually outperformed robots. Foxconn installed nets on the roofs of their workshops to prevent workers from jumping headfirst. And these same people made it so that it's now simply impossible to return an iPhone to America. No matter how perfect the AI, no matter how good the next robot is at emptying laundry from the hamper and folding it, a human will always outperform it and do it better and faster.

So, the production manager finds a room, sets up the equipment, supplies everyone with tools, and listens to comments about the equipment and logistics. But that's where his responsibilities end.

I'd even ban cameras in the assembly area, since the loss of production secrets is more important than some slacker in an office constantly monitoring who's working and who's just sitting there doing nothing.

Production secrets are what gradually build a Production Culture.

This is yet another reason why you shouldn't outsource assembly work to a third-party manufacturer.

Initially, this is seems a sound and, most importantly, simple solution.

A manufacturer with established assembly lines and supply chains should, of course, be better at this task. They already have a dedicated staff of trained assemblers. Their recruiters tirelessly trek through jungles and mountains, delving into the wildest corners of nature, just to find a village willing to work for \$10 a month.

But if you imagine the very beginning of the work, their position will be even more helpless than that of a startup. The established production culture within their environment conflicts with any innovation. Their habit of using existing solutions blocks any new perspectives.

That said, I don't consider their technology bad or backward in any way. On the contrary, I'll describe below how it's best to use their services.

But at the start, it's best to do everything yourself.



start with a clean workbench

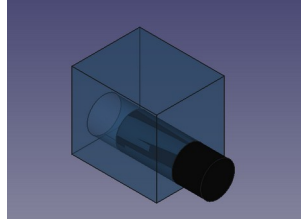
Then, when there are a lot of things on the table, it becomes equally clear that some ways of working are better than others.



However, here everything depends only on the timing.

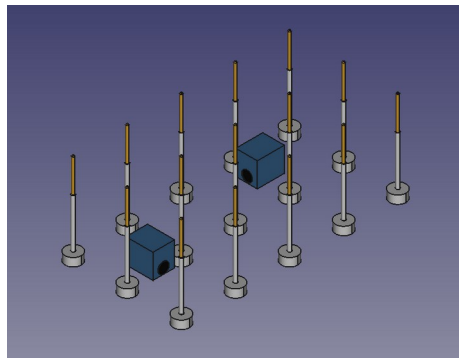
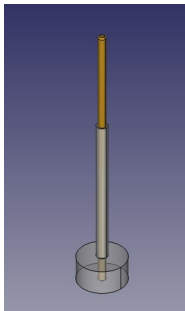
I decided to lay out all the main assembly operations in the order they occur, to roughly estimate how much time it would take initially for one person on the assembly line.

1. assembly operation



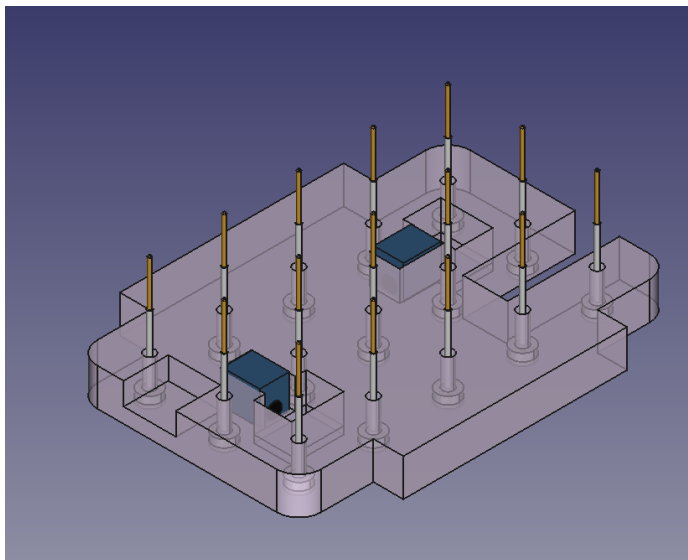
Insert the magnet into the rubber retainer. This will take a couple of seconds. This is assuming you've already sorted everything into boxes. You need six of these for each device.

2. assembly operation



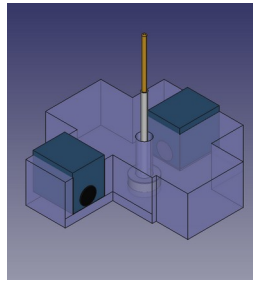
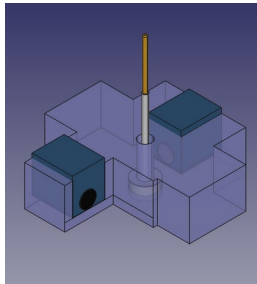
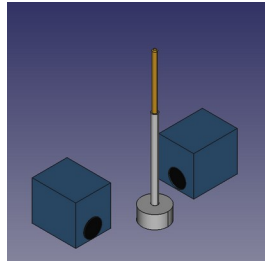
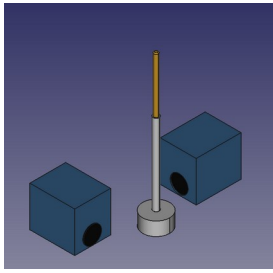
Then the next parallel operation is to take the metal sensor rod and attach the silicone shell and lower shock absorber to it. These are the main components of the moving mechanism with the sensors. They're all soft silicone rubber. But we need to prepare 17 of them.

3. assembly operation



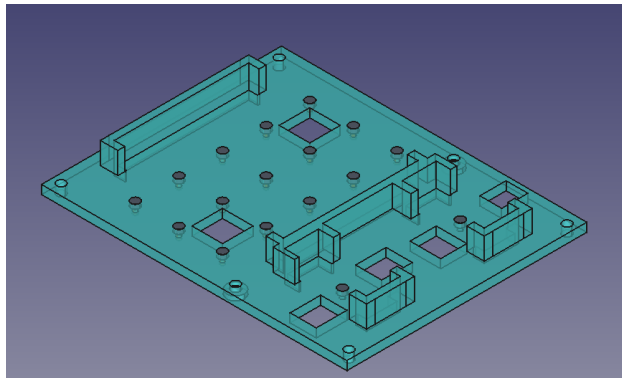
We assemble the entire moving keyboard mesh with sensors and magnets.

4. assembly operation

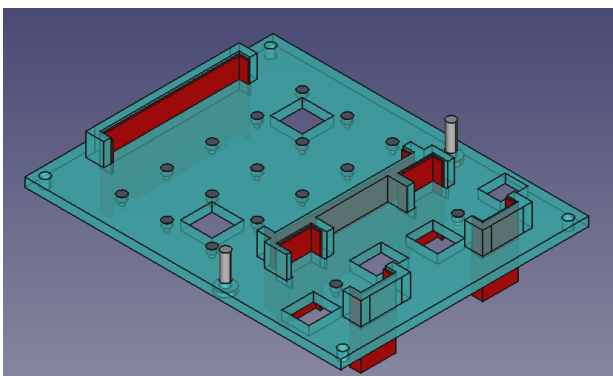


We assemble moving mechanisms with magnets and joystick sensors in the same manner.

5. assembly operation

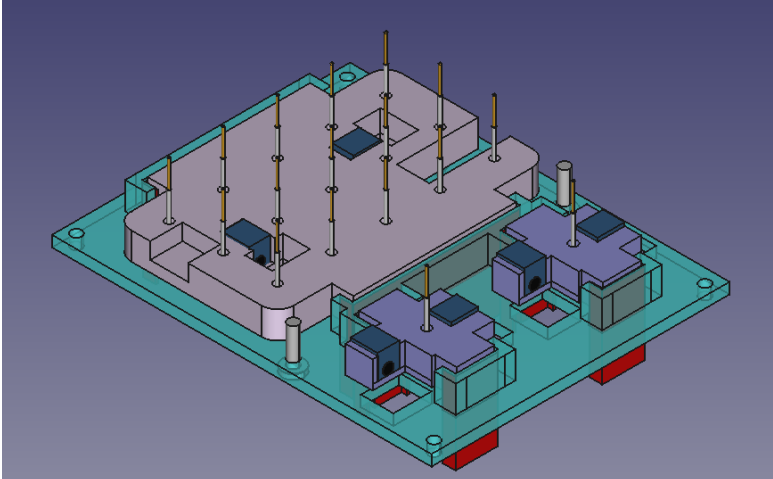


We use a mounting support stand with barriers for moving parts and contacts already filled with solder into the sockets. This operation is carried out in a muffle furnace, and the time is calculated as in the pre-production section.



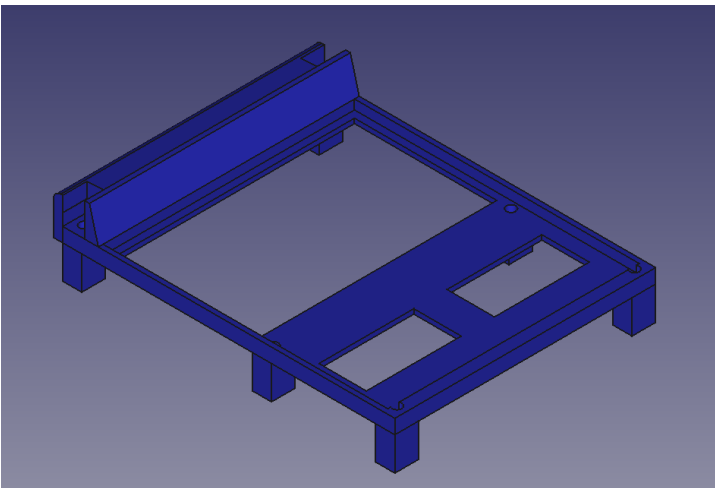
We install the clamps for the moving parts and the mounting rods.

6. assembly operation

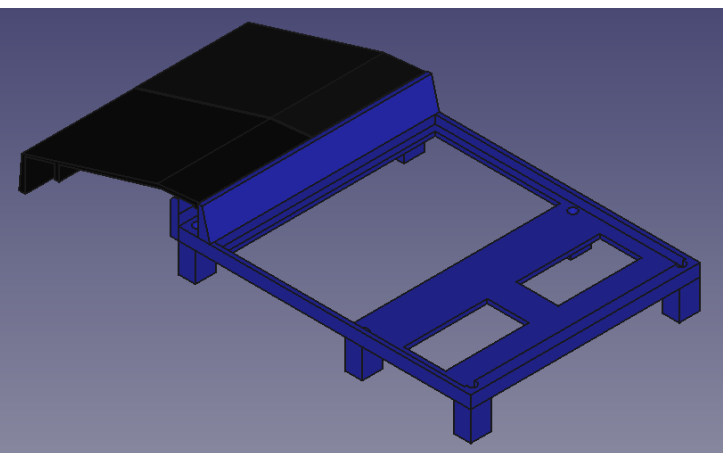


We assemble everything.

7. assembly operation

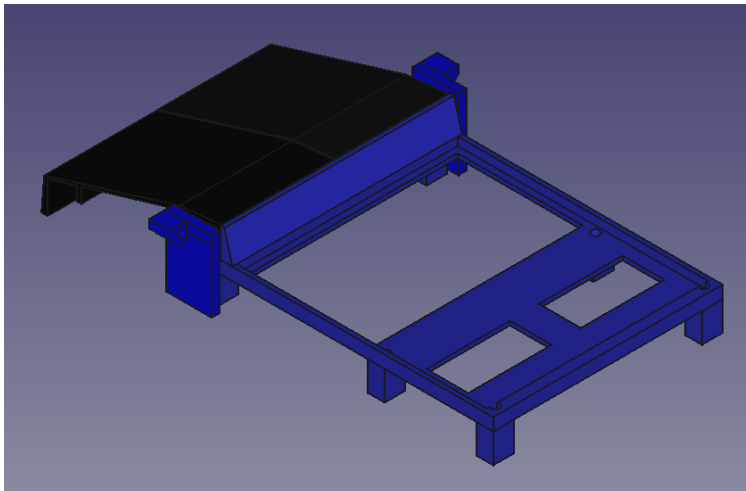


We take a basic frame with supports.



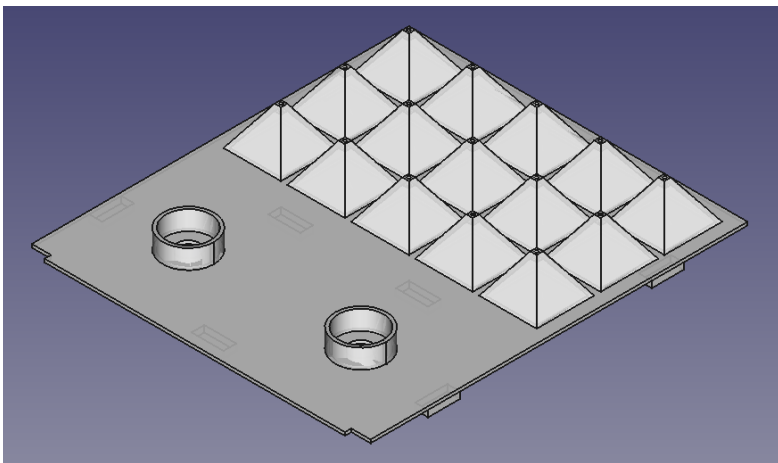
We insert mouse buttons into it.

8. assembly operation

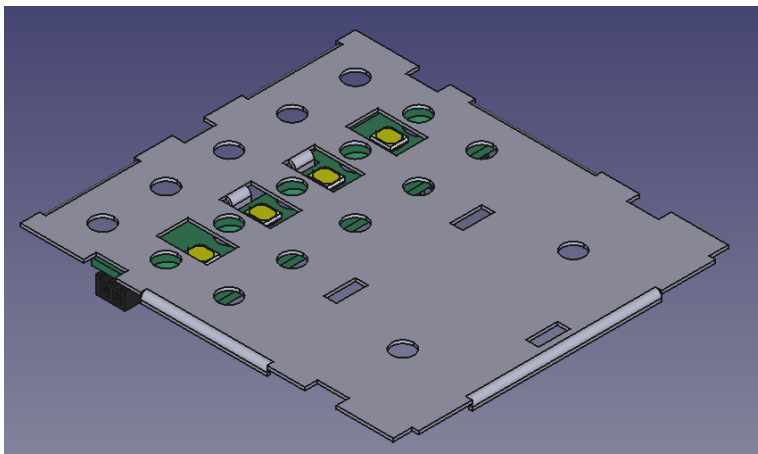


Adding side wall fasteners.

9. assembly operation

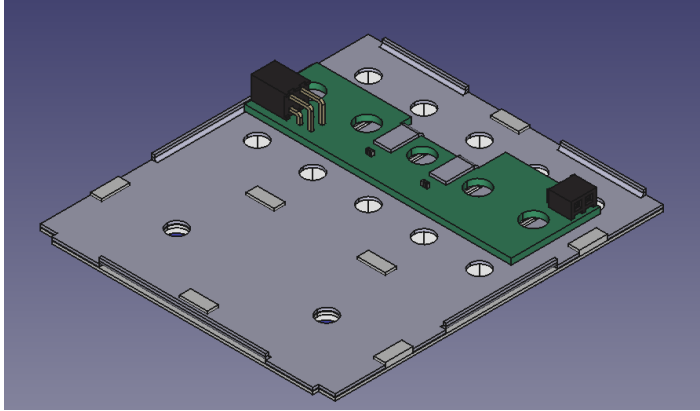


take the keyboard



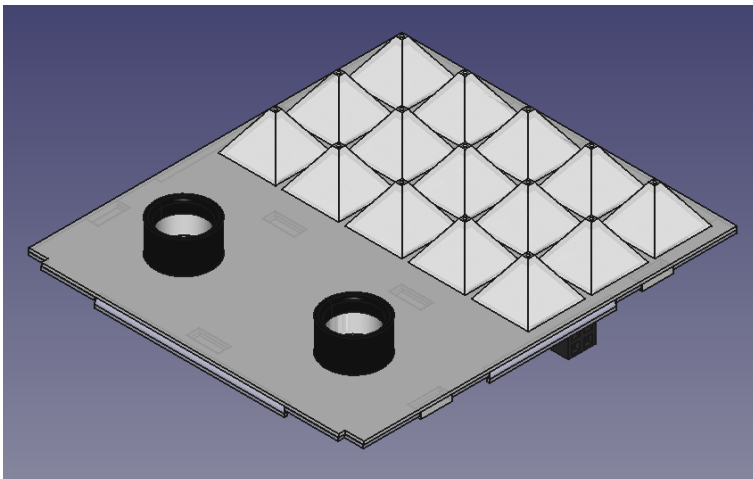
and we assemble the mounting plate with LED backlighting

10. assembly operation



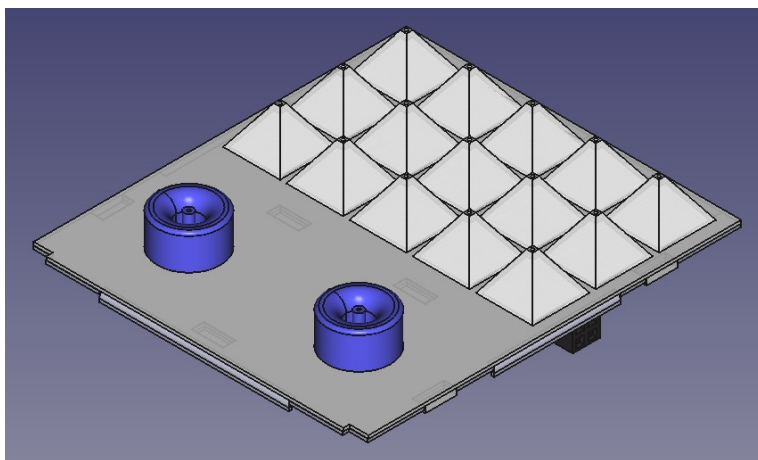
connect the keyboard to the mounting plate with LED backlighting

11. assembly operation



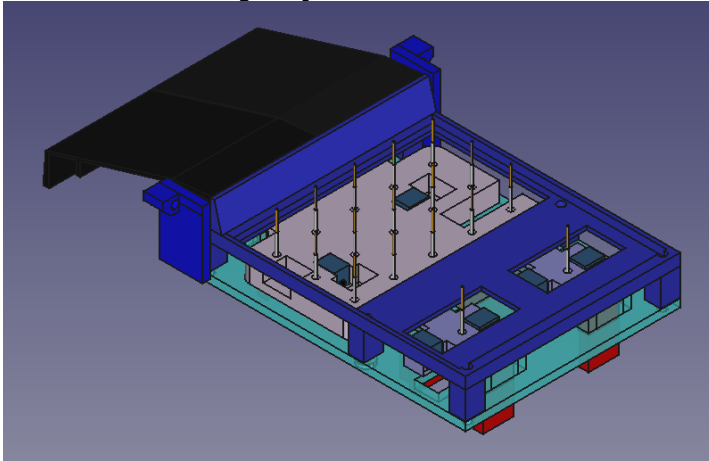
We install joystick guard rings

12. assembly operation



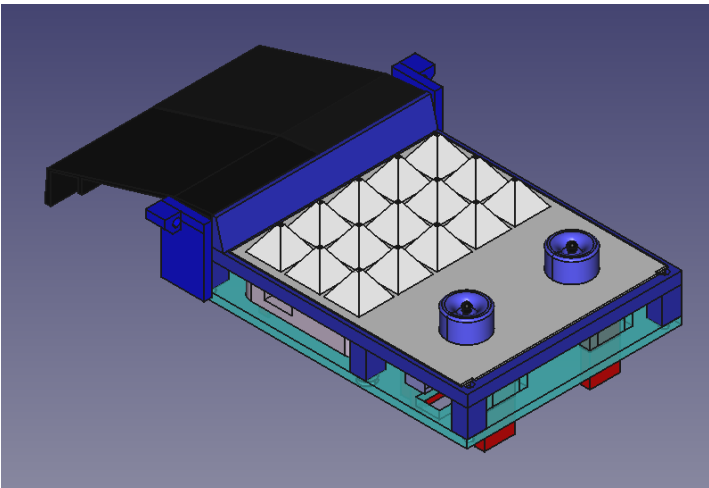
We install the joystick covers without installing the tongues or balls themselves.

13. assembly operation



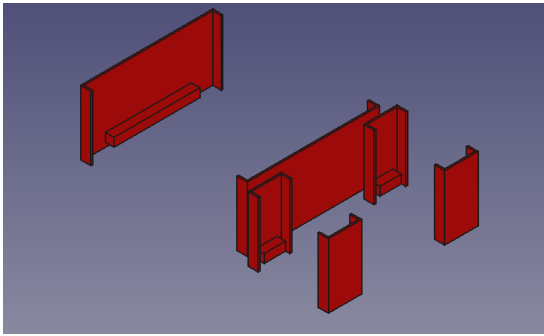
We assemble the frame of the carcass with buttons and the main moving mechanism.

14. assembly operation

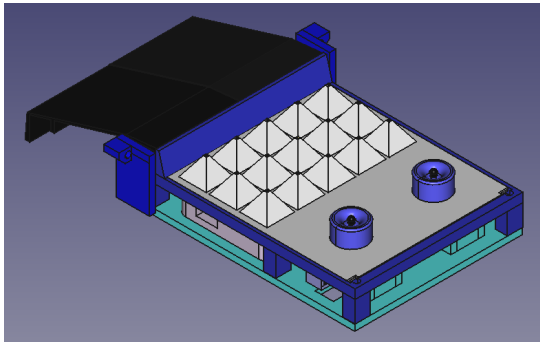


A crucial step in assembly: place the keyboard on top. Ensure the pins are positioned at the tips of the pyramids, applying pressure to the tips to secure the pin in the lower solder socket. Similarly, place the joystick balls, also applying pressure to secure the pin. Be careful not to let the assembly come apart.

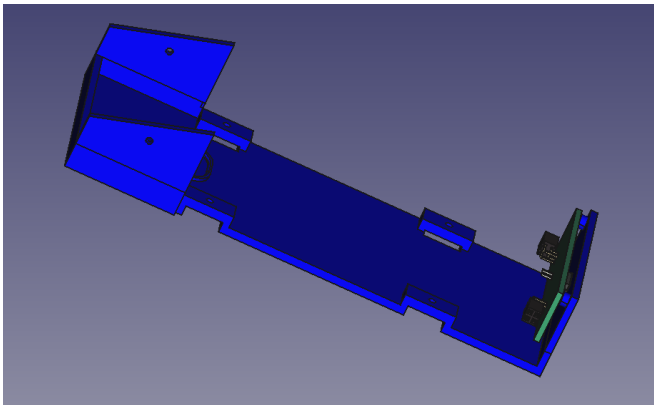
15. assembly operation



Then we remove the plugs, releasing the moving mechanism inside the assembly.

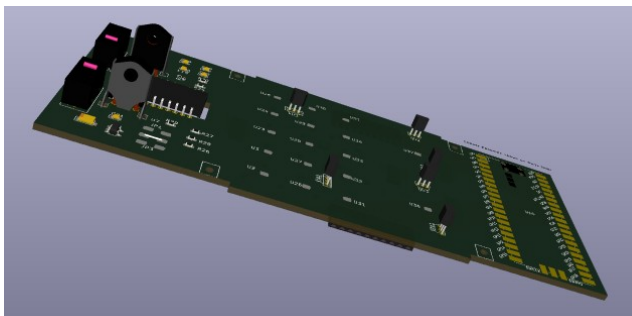


16. assembly operation

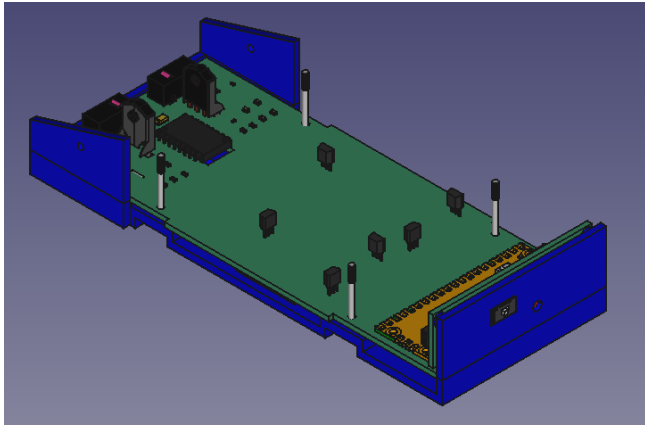


Then we connect the power backlight to the back wall of the lower base.

17. assembly operation

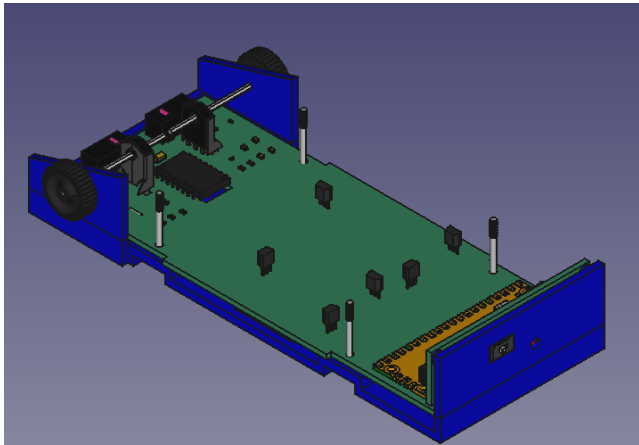


We take the motherboard



We install the motherboard on the bottom part and insert the mounting bolts.

18. assembly operation

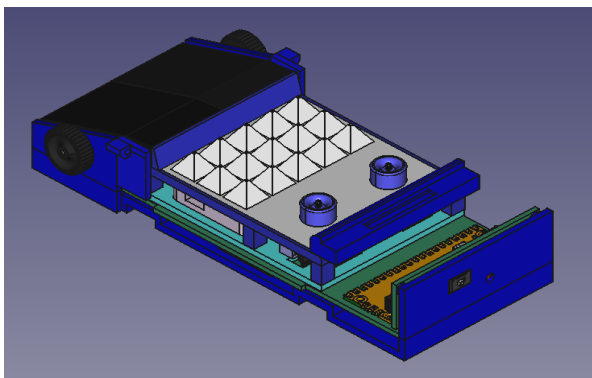


Installing the mouse axes and wheels

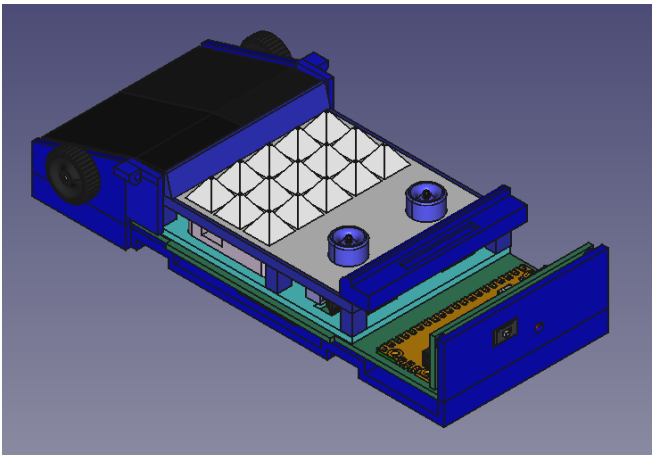
Installing the microcontroller board involves various solutions, so it's beyond the scope of this timeline.

19. assembly operation

Then we connect the upper mechanism with buttons to the lower block and the motherboard.



20. assembly operation



We are fixing the entire structure with four screws at the bottom.

This concludes the basic set of operations, which I believe should be performed independently. Further assembly concerns the side panels of the device and the power elements housed within them. I don't consider my design here to be optimal. Perhaps a third-party consumer electronics manufacturer could achieve the goals set here more effectively.

Calculating the cost of basic assembly operations

A friend of mine who worked for a long time as an assembly line foreman once told me that any realistic work quota can be completed in five hours. More precisely, the rates given for an entire eight-hour workday can be completed in five hours. Of course, standards can be raised and rates lowered, but production standards may not be met, and the level of defective work will increase, making such an increase in standards or a reduction in rates economically unviable. Many company managers resort to overtime, extending the workday to 10 hours or more. However, this leads to staff burnout and turnover.

The point is that these three hours remaining at the worker's disposal are essential for an experienced worker to establish the production environment itself. This includes preparing the workstation so that everything is convenient and within easy reach, checking logistics and communications, and a personal meditation period. It's also a stress-free timeout, when something can go wrong or go wrong.

In my cost calculations, I base the workday on an eight-hour period and the assembly day on a five-hour period. This means that in the latter case, I calculate the daily assembly rate based on the fact that the assembler's active time is five hours per day. However, overhead costs and rent are still calculated based on an eight-hour workday.

When calculating the assembly cost itself, I decided not to rely on pure timekeeping. My experiments would be imprecise, one way or another, and wouldn't provide a complete picture.

My approach here will be quite different. I've already broken down the entire assembly process into individual operations. Now I need to imagine what an assembly operation is from the worker's

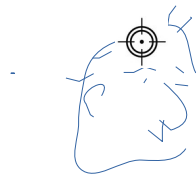
perspective. It's not a few seconds ticked off by a technologist's stopwatch, but a mental operation, where they pursue a specific goal. A person is faced with one specific goal, which they must achieve in that moment. And then, forgetting about it, they immediately move on to the next one.

This is how so-called "Quick Memory" can control human actions. It's what we just saw and immediately remembered. But if it doesn't transfer to another area of our brain, then after a while, what we saw completely disappears from our minds. The approximate maximum duration of such memory is up to 30 seconds. But I take a more reliable range from a third to half of that time. Thus, these are intervals of 10 and 15 seconds, depending on how many objects need to be retained in memory.

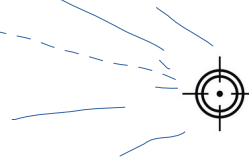
10 seconds



15 seconds



< 30 seconds



I decided that the assembler performs a production operation for as long as he can remember it. And that's consistently 10-15 seconds. This resulted in twenty and a half minutes, or 0.31 man-hours, for assembling the device (base unit). Let me remind you that this is purely manual assembly, designed for the most limited initial production run. If a team of 10 assemblers assembles a batch of 10,000 units, the assembly labor cost per unit will be \$12. Overhead costs, consisting of the salaries of all production participants, as well as rent and electricity costs, will add another \$16. Since overhead and assembly costs can easily be multiplied by a double factor, the total cost will be \$56.

Add to this the cost of components

The base is a plastic injection molded kit.

In this case, this would be the entire required batch.

A batch of 100,000 plastic parts made by injection molding would likely cost

\$170,000 and \$550,000+ in total, which breaks down to approximately **\$1.70 to \$5.50 per part**.

The exact price depends heavily on the part's design, material, and the complexity of the mold required.

Cost Breakdown

The total cost is composed of two main elements:

- **Tooling (Mold) Cost:** This is a one-time, upfront expense that generally ranges from **\$25,000 to \$100,000+** for a high-volume, hardened-steel mold. This cost is then distributed across all 100,000 parts, significantly lowering the per-unit cost compared to small batches.
- **Production Cost per Part:** This is the recurring cost for material, machine time, and labor for each individual part. For large volumes like 100,000 units, the variable cost per part can be as low as under \$1 for some simple items, with typical ranges for material (e.g., commodity plastics like polypropylene) being \$1-\$5 per pound.

Key Factors Affecting the Price

The total cost can vary widely based on several variables:

- **Part Complexity & Size:** More intricate designs with tight tolerances or undercuts require more complex and expensive molds and can increase cycle times.
- **Material Type:** Commodity plastics (like PP or ABS) are much cheaper than engineering-grade or high-performance plastics (like polycarbonate) which can cost significantly more per pound.
- **Number of Cavities:** A mold with multiple cavities produces several parts in a single cycle, which drastically increases production efficiency and lowers the cost per part, but increases the initial mold cost.
- **Secondary Processes:** Additional services like special surface finishes (e.g., painting, printing, electroplating), assembly, and specific packaging will add to the final price.
- **Manufacturer Location:** Manufacturing the parts and mold overseas (e.g., in China) can offer initial cost savings, but may lead to potential issues with quality, communication, and shipping times, which can increase the total cost of ownership in the long run.

Calculating the cost of plastic injection-molded components this way will increase the cost by another \$104 (see appendix to the financial plan). This brings the total calculated costs to almost \$160. Waiting costs should also be added. The mold production itself will likely take one to two months. Production can range from 100 units per day to several hundred, depending on the equipment load and the number of shifts that can be organized. Therefore, the wait for the first batch can extend up to five months.

Motherboard

For a two-layer, 100x50 mm board with surface-mounted components (two encoders and passive elements) and an order of 100,000 units, the estimated cost per fully assembled unit is approximately

\$1.50 to \$8.00.

Cost Breakdown

The final price is a combination of the bare PCB fabrication and the assembly cost, which includes components and labor.

- **Bare PCB Fabrication Cost:** For a simple 2-layer board of this size (approx. 7.75 sq in) in high volume, the cost is likely to be on the lower end of the mass production range, estimated between **\$0.50 and \$1.50 per unit**.
- **Component & Assembly Cost:** The cost for the two encoders and passive elements (resistors, capacitors, etc.) will largely determine the final price.
 - **Components:** Passive components are inexpensive in bulk, but the encoders can vary in price from a few cents to a few dollars each depending on their type and quality.
 - **Assembly (SMT Labor):** Surface mounting technology (SMT) is highly automated, which helps reduce cost per board in large volumes.
- **Total Assembled Cost (PCBA):** Considering all elements, the total cost for a finished board is estimated at **\$1.50 to \$8.00 per unit**.

Key Cost Factors

- **Component Selection:** The specific model and quality of the two encoders are the biggest variable cost factors.
- **Material:** The use of standard FR4 material keeps costs low.
- **Design Optimization:** For mass production, efficiency in panel utilization during manufacturing can help minimize material waste and cost per board.
- **Testing:** Depending on the testing requirements (e.g., AOI, functional testing), an additional cost per unit might be incurred.

Adding an optical mouse sensor, two mouse buttons, and a touch sensor to your board for a 100,000-unit order would increase the estimated cost per unit to approximately

\$2.00 to \$12.00.

Adding a

Raspberry Pi Pico to the board for a 100,000-unit order would increase the total cost by approximately **\$3.60 to \$4.00 per unit** compared to the original, non-microcontroller board.

Cost of Incorporating a Raspberry Pi Pico

The original Raspberry Pi Pico board typically retails for \$4 in single units. In a mass-production scenario, you have two options that affect pricing differently:

Option 1: Using the Pre-made Raspberry Pi Pico Module

If you directly integrate the complete Raspberry Pi Pico module onto your board design, you're looking at a cost increase of around **\$3.60 to \$4.00 per unit** in bulk quantities.

- **Pros:**
 - Saves significant engineering time by using a pre-certified module.
 - Simplifies manufacturing and testing processes.
- **Cons:**

- Slightly larger physical footprint compared to a bare chip solution.
- Higher cost per unit than designing a custom circuit with the bare microcontroller.
-

Option 2: Designing the Circuit with the Bare RP2040 Chip

A more cost-effective approach for a 100,000-unit order is to design your custom PCB to incorporate the bare RP2040 microcontroller chip itself, rather than the entire Pico module.

- **RP2040 Chip Cost:** The bare RP2040 chip costs approximately **\$0.70 per unit** when purchased in large reels (e.g., 3,400+ units).
- **Additional Necessary Components:** To function, the bare RP2040 chip requires external QSPI Flash memory (to store the program code) and some supporting circuitry. This typically adds another **\$0.10 to \$0.20 per unit** for the flash memory.
- **Total Bare Chip Cost:** Approximately **\$0.80 to \$0.90 per unit** (plus a minor increase for extra SMT points/assembly).

By using the bare chip, you save roughly \$3 per unit compared to using the full Pico module. This results in a total cost per unit of the entire assembled board with all components (encoders, mouse parts, etc.) of approximately **\$2.30 to \$9.00**

This will increase the cost price to approximately \$170-180, which, together with the percentage charged by the trading platform plus other expenses, will be at the level of the planned selling price of \$200 (199).

Unstructured area — the supplier will not be ready

How willing will the component supplier be to provide such a loan for tooling for the first trial batch? The difference between the planned 100,000 and the first batch of 10,000 means the supplier can expect to receive 10% of the total order amount? This is despite the fact that the supplier will have to manufacture all the working equipment - molds – himself.

The only option left is to abandon small trial batches and immediately try to raise funds for a full batch of 100,000 units, or at least half that amount. This will, of course, again create assembly problems. The number of assemblers required will now be an order of magnitude greater. The overall complexity of the production process, which still needs to be organized, will also increase. It's quite possible that the commitments made will not be met. The solution would then be to increase the selling price by approximately two or three times, to \$400-\$600. However, it seems this will be fruitless, as none of the stated goals will be met. It will be impossible to reach the 100,000 unit quantity, and what has been raised will not be enough.

Considering the alternative, we can assume that the supplier will agree, but will put these works in the second order of execution. This will inevitably lead to delivery delays and missed deadlines. And then buyers on the platform habitually wait and complain, overhead costs will skyrocket at this time.

Ultimately, the best option will be to sell the company.

Structured area - basic idea

Mold design should begin immediately. The supply chain manager shouldn't be involved in this matter. This way, once hired, they can get to work immediately and there won't be any downtime. Before that, there will be the initial stage of burnout.

During this period, the following will be completed:

- company organization
- company opening
- staff hiring (2-3 months)
- work organization (1 month)
- initial stages of work - firmware and electronics (2 months)
- testing the initial stages of work - one week
- software writing (2-3 months)
- creating the first prototype devices - one week
- creating tooling drawings - one month
- creating mold drawings - one month

If all these processes are properly organized, the total time for the first period can range from 3-4 months to six months.

The creation of the promotional video should begin with sending out invitations. This will be an invitation to participate in testing the new device. Those who express interest can be sent pre-production units of the device, which will later provide material for the promotional video. Obviously, the number of people who receive these invitations and subsequent copies will differ from those who actually interview. It's important to anticipate that some recipients will ignore the mailing. To make this process more effective, a questionnaire should precede the mailing. We need to gather a base of people who might be interested. These could be media representatives and developers who regularly have to deal with writing. They could also be people who prefer to work outside the home or travel frequently. This stage can also take a couple of months. It's also important to keep in mind that interviewees also need time to get used to the new device. After creating a promotional video with an interview, you can begin raising funds on a crowdfunding platform. The typical timeframe is about a month.

It's worth recalling the basic idea of crowdfunding: the buyer is an investor and therefore willing to wait for their money to be converted into a finished device. Since the price of 199 assumes a 10% margin, I can confidently say the actual selling price will be higher. Therefore, after finishing sales on this platform, I reserve the right to sell at a higher price, say, on Amazon. This should seem like an additional argument for the buyer on the platform that it's worth buying now.

10,000 units isn't a specific goal, but a starting point. Having calculated the initial cost of 10,000 units, you can scale up from there. It's also important to remember that the company's subsequent valuation will depend on how this initial quantity is sold. This will also provide an opportunity to raise additional capital. While a halt in sales is desirable, the scale can vary greatly. It could be 15,000 or 20,000. It's important to understand how sales will perform.

I've already calculated that producing a batch of 10,000 devices requires ten assembly workers and 69 eight-hour workdays.

However, delays in tooling production by the supplier, production itself, and logistics must be added to this timeframe.

We're ordering approximately 20 unique plastic injection molded parts, each requiring its own mold. These aren't very large molds—I'd even call them small—and for simplicity, we can assume the cost of each will be around \$60,000, of which the supplier will require 30% as a down payment. This amounts to \$360,000. The supplier will also likely require a third as an advance payment for the first batch of products, which would be another \$312,000. The total cost, therefore, should be \$672,000.

After paying all this, we'll only have to wait perhaps two months.

Rent and electricity

The average cost to rent industrial space in the US varies significantly by location, but recent data (late 2025) suggests a national average around

\$8.73 to \$9.30 per square foot annually, translating roughly to **\$0.73 to \$0.78 per square meter per month**, though this is usually quoted monthly as base rate plus expenses (NNN) for properties often costing around **\$1.10 per square foot per month total** in various markets, with coastal/major hubs like NYC/LA much higher and inland areas lower.

Key Factors Influencing Cost

- **Location:** Major ports (LA, NY) and logistics hubs command higher rents (\$18-\$35+/sq ft/yr) than growing Texas markets (\$8-\$16/sq ft/yr).
- **Market Type:** Industrial (warehousing/logistics) is often quoted monthly (e.g., \$0.85 base + \$0.25 NNN) versus yearly for offices/retail.
- **Lease Type:** Most are Triple Net (NNN), meaning tenants pay base rent plus taxes, insurance, and maintenance.

Examples of Average Rates (Per Square Foot Annually)

- **National Average:** Around \$8.73 - \$9.30.
- **New York:** \$25 - \$50.
- **Los Angeles:** \$18 - \$35.
- **Dallas:** \$8 - \$16.

Conversion to Square Meter (approximate)

- 1 square foot (sq ft) $\approx$ 0.093 square meters (sq m).
- An average rate of \$9.00/sq ft/year is roughly $\$9.00 / 0.093 \approx \96.77 per square meter per year, or about **\$8.00 per square meter per month**.

In summary, expect to see prices quoted per square foot (monthly or annually) with NNN, but a rough estimate for an average US location is around \$1.00 - \$1.20 per sq ft per month total, or roughly \$0.75 - \$0.90 per square meter monthly, depending heavily on the specific city and building quality.

Thus, if you rent an area of 400 square meters for a year, the average rental cost will be \$4,500.

Electricity costs in the U.S. average around

17-19 cents per kilowatt-hour (kWh), leading to an average monthly bill of roughly **\$140-\$170**, but prices vary significantly by state, from cheaper states like [Kentucky](#) to expensive ones like [Hawaii](#), due to generation mix, grid costs, policies, and climate.

National Averages (Late 2025/Early 2026 Estimates):

- **Average Rate:** Around 17.5¢ to 18.9¢ per kWh.

- **Average Monthly Bill:** Roughly \$140 to \$170.
- **Average Annual Usage:** About 850-900 kWh per month.

Factors Influencing Costs:

- **State Policies & Regulations:** Each state sets its own energy rules, impacting prices.
- **Power Generation Mix:** Costs depend on using fossil fuels (coal, gas) or renewables (solar, wind, nuclear).
- **Grid Infrastructure:** Transmission and distribution costs vary by terrain and grid age.
- **Climate:** Heating and cooling needs significantly affect usage and bills.

Examples of State Variations (Late 2025 Data):

- **Higher Costs:** Hawaii (\$0.40/kWh), [California](#) (\$0.30/kWh), [Massachusetts](#) (\$0.27/kWh).
- **Lower Costs:** [North Dakota](#) (\$0.10/kWh), [Kentucky](#) & [Tennessee](#) (\$0.12/kWh).

For your specific cost, check data from the [Energy Information Administration \(EIA\)](#) or resources like [ElectricityPlans](#) and [SolarReviews](#) for state-by-state breakdowns.

To estimate the annual energy consumption for an electronic equipment assembly shop with 40 employees and 400 square meters, we calculate based on industry-standard intensity metrics for light manufacturing.

Step 1: Calculate total annual labor hours

Assuming a standard full-time schedule of 40 hours per week for 52 weeks a year, the total annual labor hours for the staff is calculated as:

Step 2: Estimate energy consumption using labor intensity

For electronics assembly (a low-intensity sector), the energy use is approximately 64 MJ of primary energy per worker-hour.

Converting Megajoules to kilowatt-hours

Step 3: Estimate energy consumption using floor area intensity

Electronics assembly shops typically have an electricity intensity of 150 to 300 kWh/m² annually, covering lighting, HVAC, and bench equipment (soldering stations, testing rigs).

- Lower Bound:
- Upper Bound:

Answer:

For an electronic assembly shop of this scale, the actual site energy consumption is estimated to be between 80,000 and 150,000 kWh per year. While labor-based primary energy metrics suggest higher figures (approx. 1.48 GWh) by accounting for the total energy supply chain, the operational utility demand for a

facility typically falls in the lower range due to the compact footprint and the relatively low power draw of assembly tools compared to heavy machinery.

Unstructured area - problem of hiring employees

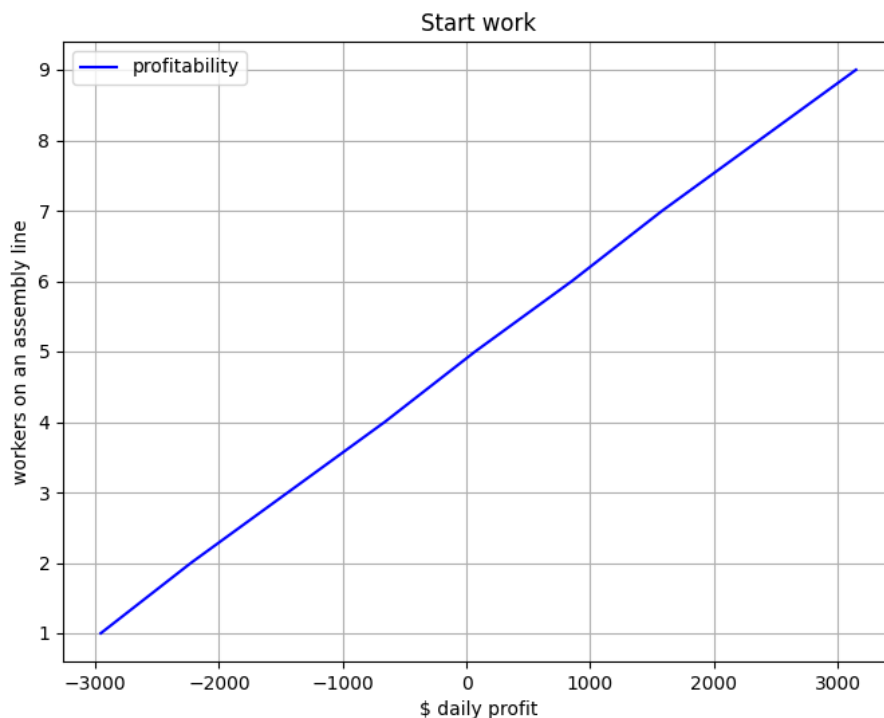
It is important to understand right away that people are not machines, and everyone works - some faster, some slower. And the conveyor belt cannot work faster than the slowest worker. Low margins allow for lower sales prices, but there is always the risk that profits will go into the red.

Structured area

If the minimum fundraising for the first company reaches \$2 million, or 10,000 orders, the company can be launched. The down payment for molds could be around \$360,000, and the advance payment for the first batch of components could be around \$312,000. For motherboards and electronics, the entire order can be paid for upfront. It's also worth considering the cost of initial downtime, when staff has been hired and space has been rented, but work will only begin once equipment is installed and components arrive. Under other circumstances, everything will be ready to begin, but the assembly line will be understaffed.

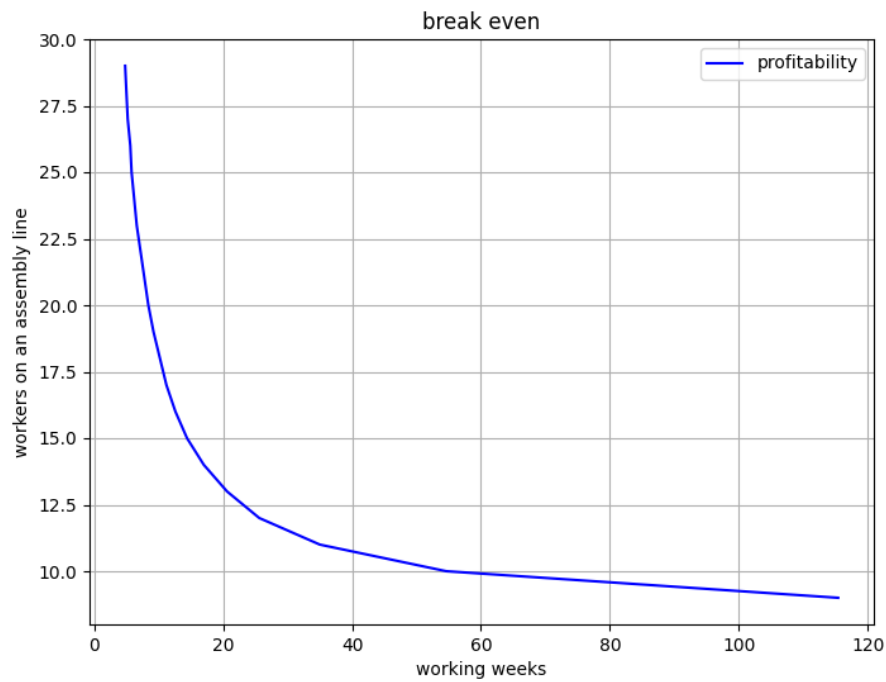
What's the minimum number of employees needed to ensure daily operating profit doesn't slip into the red? Here's a rough start-up schedule.

I take an overhead level of around 50% as a possible starting point, but it can then jump to 150% due to various unfavorable conditions and downtime. But then it drops back to 50%.

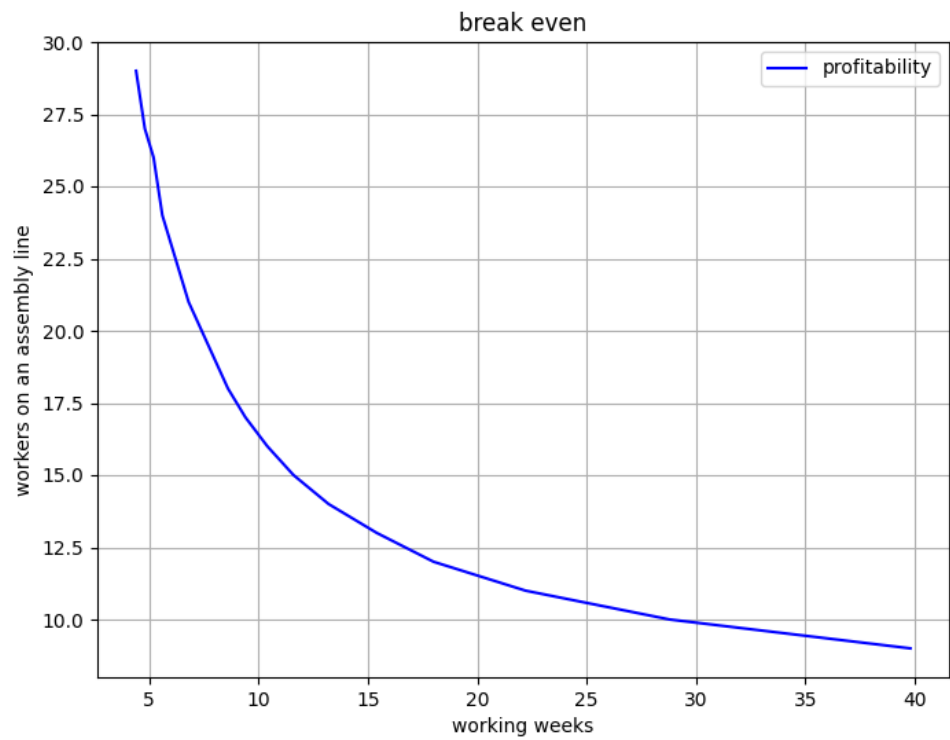


Work can begin when the assembly line reaches 6-7 people. However, this number will not provide a sustainable profit or break-even point. What I mean here is the return on investment and reaching net profit. The following graphs show this possibility and timeframes for different overhead levels.

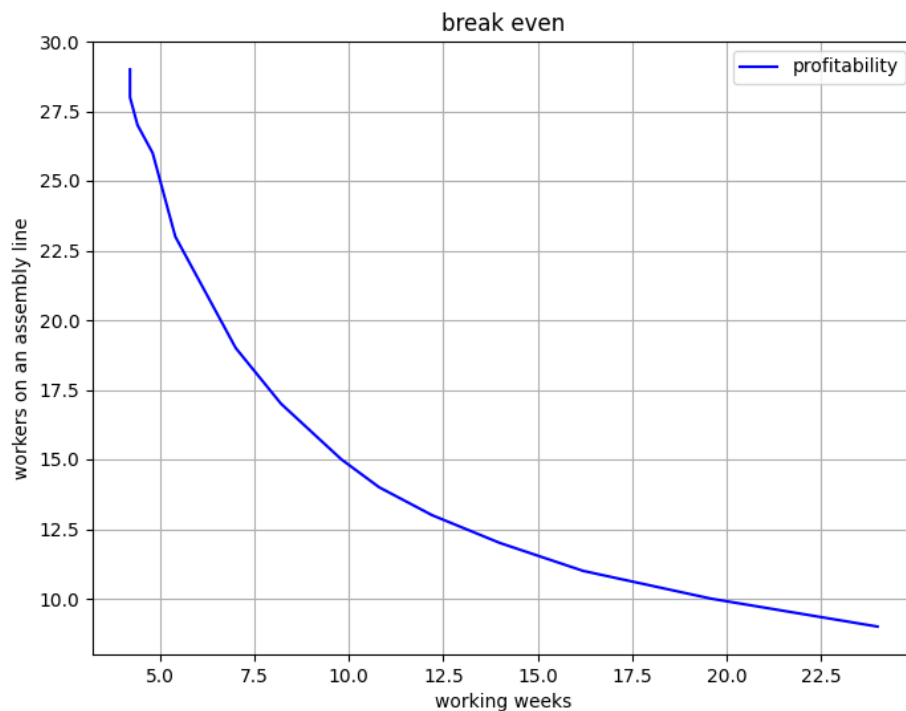
1 Overhead 150%



2 Overhead 100%



3 Overhead 50%

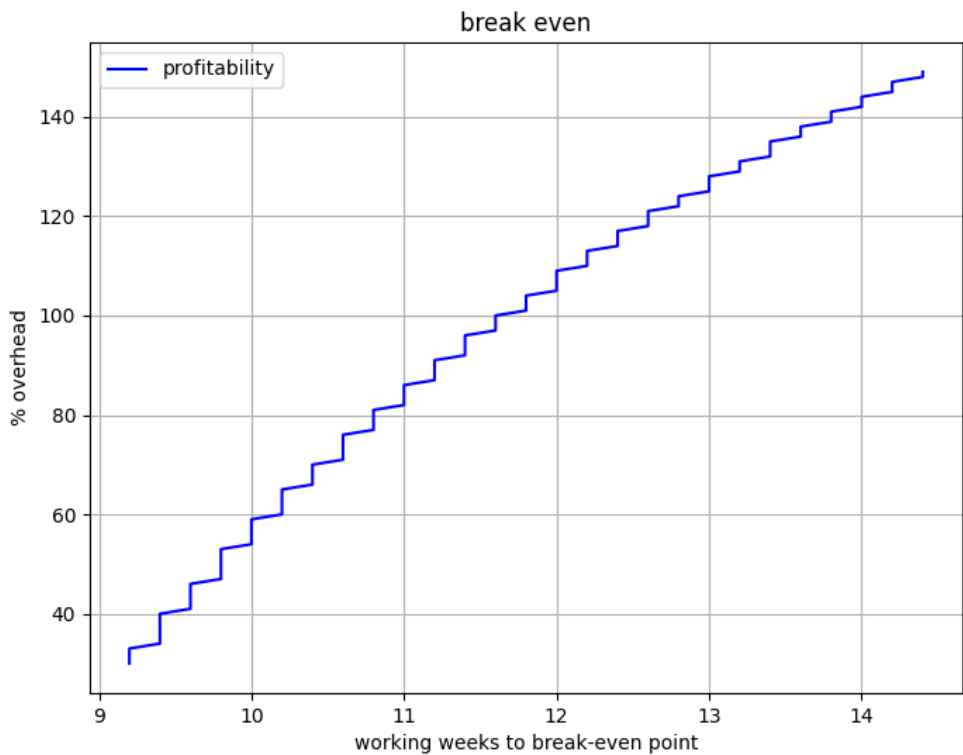


Although the graphs clearly show how increasing assembly staff leads to shorter lead times, I decided to add some additional conditions. This will obviously affect the hiring process itself. I decided that the hiring conditions would allow me to interview, on average, no more than one employee per day, as this is a complex and responsible procedure with all the paperwork and interviews. And now the picture is immediately more balanced. Hiring an additional 20 workers amounts to almost a month's work. But at the same time, constantly replenishing the assembly line without stopping work is a bad idea. This will create chaos and a host of new problems. Therefore, it makes sense to start work as early as possible and keep the workforce small.

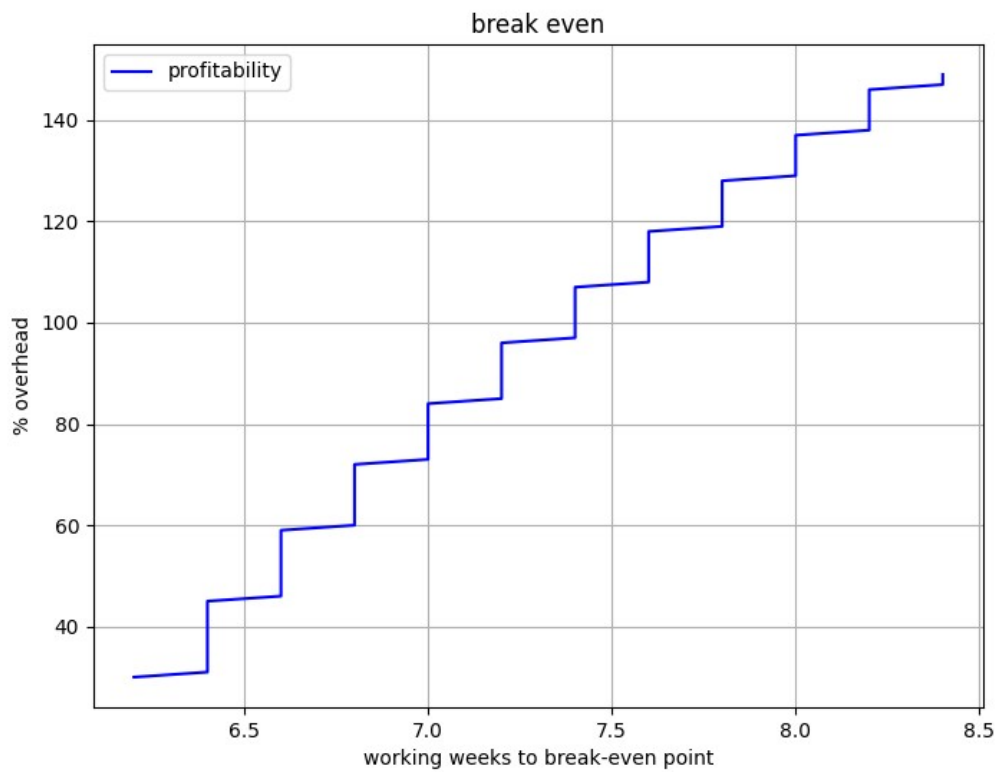
A possible option here is to employ a maximum of 15-20 people in assembly during the initial period of work. That is, it's best to limit the number to approximately this number, as further uncontrolled staffing expansion and extensive growth will only lead to interruptions and increased overhead costs. Conversely, by limiting the number of employees to such a small one, the best step would be to try to reduce overhead costs. This should also lead to an increase in production speed and volume, but without increasing the number of employees.

Here's an approximate graph of the growth this could lead to.

In the first graph, the hiring growth will plateau at 15 people



and in the second, at 20.



Thus, it can be said with confidence that when constructing a time schedule for the first year of work, the ultimate goal will be to reach the break-even point with a return on investment.

Timeline

First year

First quarter

Company organization, hiring employees, and organizing, with their help, the refinement of the device prototype into pre-production units.

1. Company organization
2. Premises and equipment
3. Ordering components
4. Hiring the following employees:
 - Design - 1 employee
 - Motherboard, electronics, firmware, and software - 1 employee
 - Firmware applications, software, and additional applications - 1 employee

Second quarter

1. Server installation, domain setup, website creation
2. Development and creation of prototypes
3. Hiring new employees:
 - Designer with a focus on technology - 1 employee
 - PR agent with interview capabilities - 1 employee

Preparation for crowdfunding

- video creation
- advertising campaign

Determining the order completion timeframe (around one year)

Third quarter

Crowdfunding fundraising

Order counting

Assembly area rental

Ordering components and equipment

Staffing

- Supply chain manager — 1 employee
- Production manager — 1 employee
- Units — 2 employees
- Assembly line workers with training — 6 or more employees

Fourth quarter

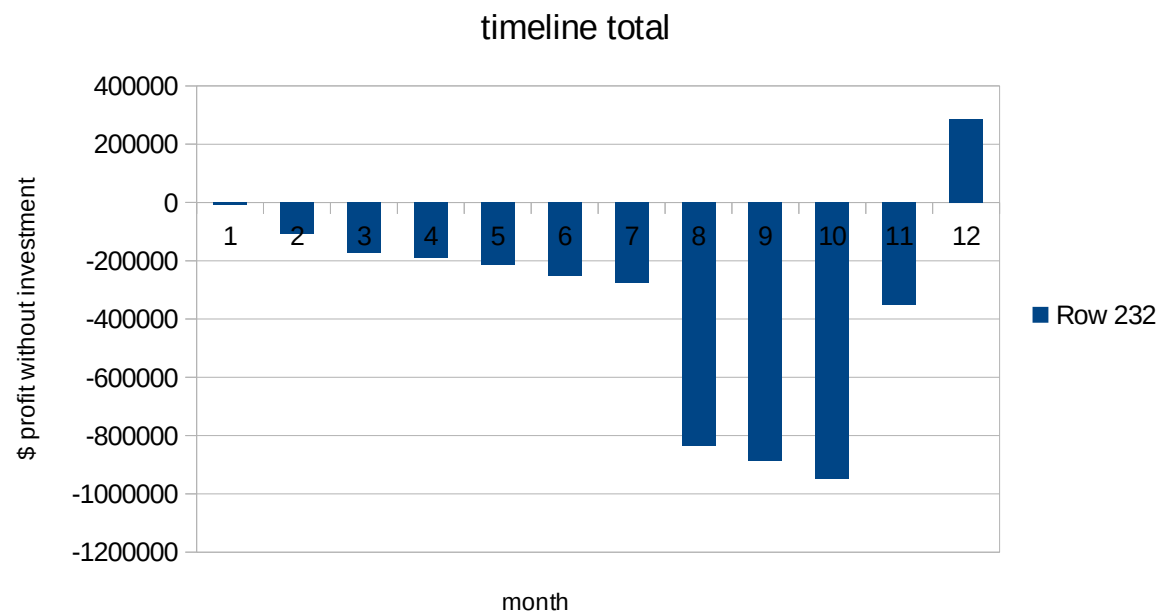
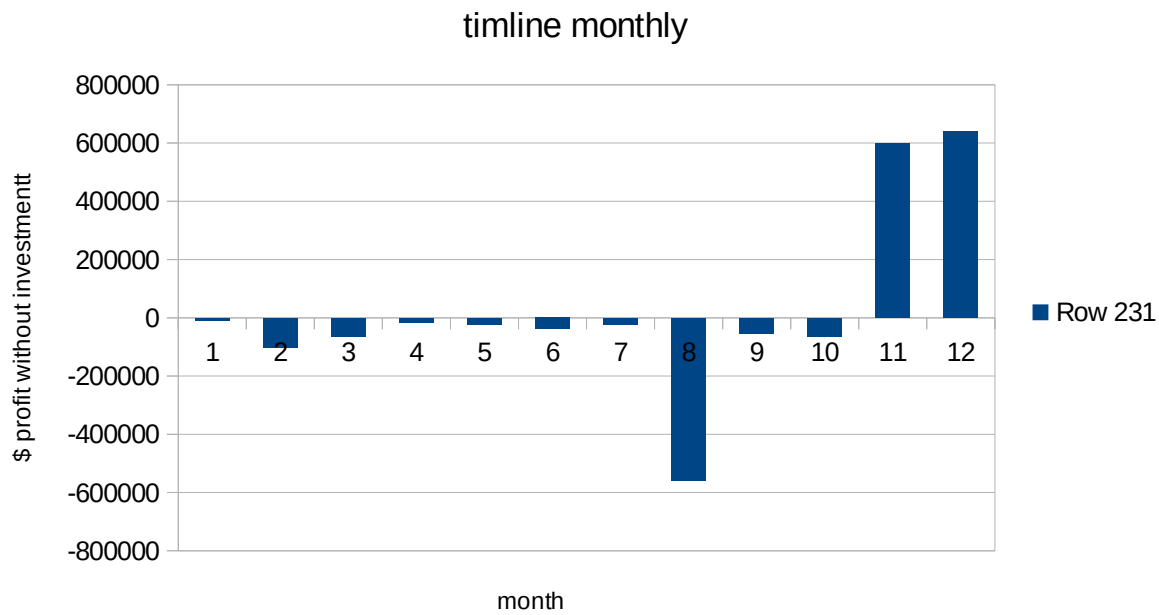
commencement of work after possible initial downtime

creation and testing of a trial batch

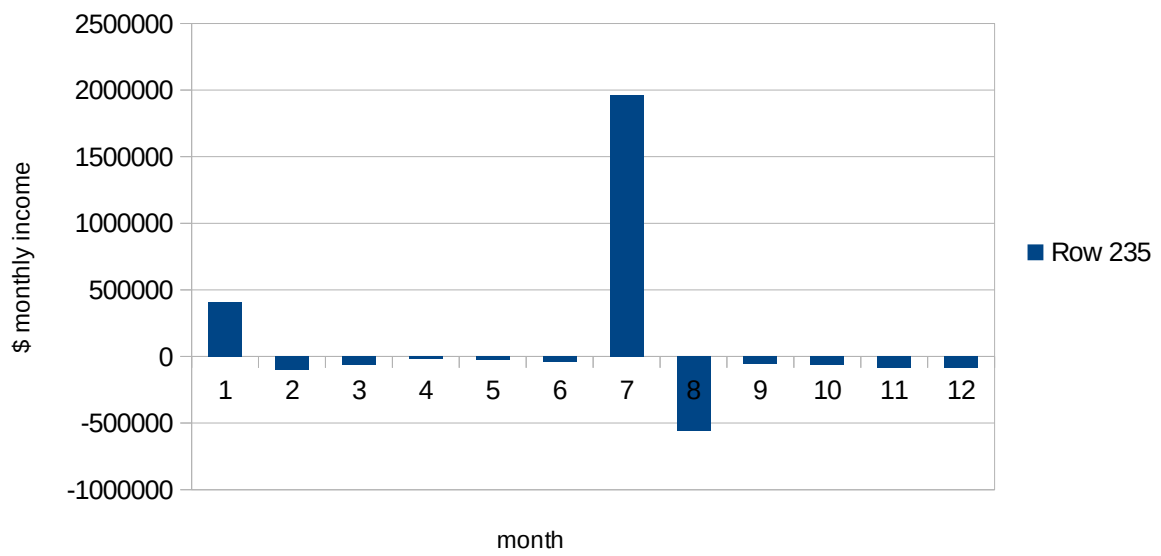
additional hiring of up to 15-20 assembly personnel

commencement of main work

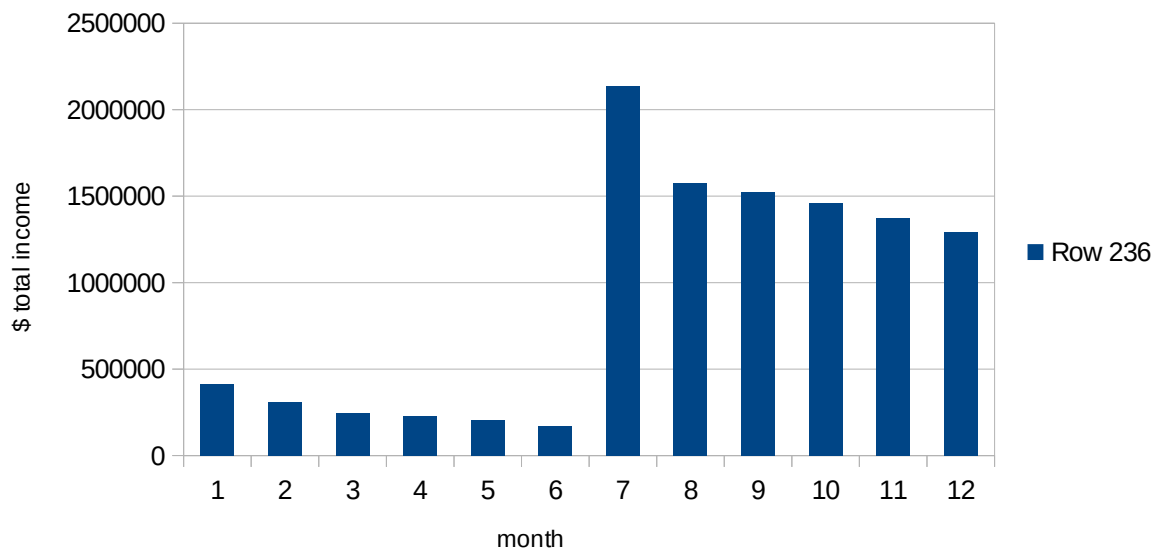
reaching the break-even point.



timeline monthly with investments



Total timeline income from investments



Second Year

Unstructured area - who will read it?

What can we realistically plan for next year if the first hasn't even begun? If everything we planned doesn't happen in the first year, what can we talk about next year? What guarantee is there that anything planned will come to fruition? But even if we get started, development will likely take a completely different path, meaning all plans for next year will be meaningless.

So the question arises: who will read this now? They'll simply say to themselves, "Yes, it can all be described that way. How grandiose it will be." But that's how various business plans were written. And most of what was written remained on paper. Everything turned out differently, not as planned. It's impossible to foresee everything, and the market situation changes, for the most part, in unpredictable ways. So all previously conceived plans go out the window.

Structuring area

By the end of the first year, the team's total size is clearly projected to be 30-40 people. The production area is likely 400 square meters, and the primary method of work is manual assembly. Approximately 5,000-7,000 units will be produced per month. The initial target of 100,000 units will only be 10% complete, as work will only begin in the final two months. At this point, the existing production resources will be reaching their limits.

I'll still need to write a plan for the second year, since the first is unfinished and incomplete without it.

What should the market look like in the meantime?

If the initial consumer to be targeted in the first year should be, to one degree or another, the developer of the new product, then in the second year one should obviously expect the product itself. The initial intended buyer is the crowdfunding participant who will receive their very first unit of the device.

It's clear that, having received this device with its new capabilities, he'll begin developing his own project. This will likely involve integration with augmented reality devices. What's the market for such devices like this?

Large companies like Apple, Google, and Facebook have already released their own solutions.

Google's primary AR device is their new [Android XR smart glasses](#), designed for all-day wear with Gemini AI for real-time translation, visual search, and AI assistance, working alongside a phone for processing; they also have the [ARCore SDK for developers](#), a software platform enabling AR experiences on many phones and headsets. While ARCore powers apps, the new glasses are Google's dedicated hardware for blending digital info with the real world.

Key Aspects of Google's AR Hardware & Software:

- [Android XR Glasses \(Upcoming Hardware\):](#)
 - **Features:** Sleek, lightweight design, integrated camera/mic, speakers, optional in-lens display, and powerful Gemini AI for real-time translation and contextual understanding.

- **Functionality:** Aims to provide hands-free AI assistance, memory recall, and enhanced navigation, working with a connected phone.
- **Partnerships:** Eyewear partners like Gentle Monster and Warby Parker are building glasses on the Android XR platform.
- **ARCore (Software Platform):**
 - **Purpose:** A software development kit (SDK) for building AR apps that run on compatible Android phones and other devices.
 - **Capabilities:** Uses motion tracking, environmental understanding (surfaces, light), and depth sensing to place virtual objects realistically in the real world.
- **Android XR (New OS):**
 - This is the new operating system for "extended reality" devices, including future headsets and smart glasses, extending Android's reach into AR/VR.

In essence, Google is moving from AR software (ARCore) to dedicated, AI-powered hardware (Android XR Glasses) to deliver immersive AR experiences directly through eyewear, powered by the Android XR platform.

Apple's primary AR device is the [Apple Vision Pro](#), a high-end spatial computer that blends digital content with the real world, functioning as a mixed reality (MR) headset for augmented reality (AR) and virtual reality (VR) experiences, using cameras for "video see-through" to overlay virtual elements onto your view of the physical world. It's Apple's first major new product category in years, marketed as a "spatial computer" for running apps and consuming media in a 3D environment.

Key Features & Functionality:

- **Spatial Computing:** Blends digital apps and content seamlessly with your physical surroundings.
- **Video See-Through:** Uses external cameras to capture your environment and display it on internal screens, allowing for AR overlays without traditional see-through lenses.
- **Control:** Operated with your eyes, hands (finger gestures), and voice, with a Digital Crown for adjusting immersion.
- **Hardware:** Features an M2 chip, an R1 spatial co-processor, high-resolution micro-OLED displays, and LiDAR scanners.
- **VisionOS:** Runs on a new operating system designed for spatial computing, integrating with Apple's ecosystem.
- **Mixed Reality:** Offers both AR (blending digital with real) and full VR (immersive virtual environments) experiences.

How it Achieves AR:

Instead of transparent lenses, the Vision Pro uses high-resolution cameras to constantly scan the user's surroundings, creating a live digital feed. Virtual objects and apps are then rendered onto this feed, creating a convincing augmented reality experience

Facebook's primary, true augmented reality (AR) device unveiled for the future is called **Orion** (formerly codenamed Project Nazare). Unveiled in September 2024, these are described as the most advanced pair of AR glasses ever made, combining the look of regular glasses with fully immersive AR capabilities.

- **Orion AR Glasses:** These glasses use holographic displays to overlay virtual images on the real world, utilizing Micro LED projectors, silicon carbide lenses, and specialized custom silicon to manage power and weight. They are designed to be used with a wireless "puck" (compute unit) and a neural wristband for control.
- **Neural Wristband:** Developed by Reality Labs, this wearable device uses electromyography (EMG) to interpret nerve signals from the brain to the hand, allowing users to control AR interfaces with small finger movements.
- **Project Aria:** A research initiative (not a consumer product) that uses sensor-rich glasses to capture data (video, audio, eye movement, location) to train AI systems for future AR, navigation, and mapping.
- **Ray-Ban Meta Smart Glasses:** While these are technically "smart glasses" with limited AI capabilities (camera, audio, voice assistant) rather than full AR, they are part of Meta's hardware ecosystem developed in partnership with EssilorLuxottica.

Orion is currently available to Meta employees and select audiences to refine the technology for a future consumer product line.

Besides Meta (Facebook), Apple, and Google, the augmented reality (AR) hardware market is driven by a mix of specialized enterprise companies and consumer electronics manufacturers. As of 2025–2026, the key producers of AR devices, including mixed reality (MR) headsets and smart glasses, are:

Major AR Hardware Manufacturers

- **Microsoft:** Known for the **HoloLens 2**, a leading AR headset for enterprise, manufacturing, and medical applications.
- **Magic Leap:** Focuses on enterprise AR with the **Magic Leap 2**, offering high-quality, lightweight headsets designed for industrial use.
- **XREAL (formerly Nreal):** A leader in consumer-friendly AR glasses (e.g., XREAL Air 2 Pro), with devices that connect to phones and computers for a "virtual screen" experience.
- **Vuzix:** Produces smart glasses primarily for industrial, warehouse, and remote support applications.
- **Lenovo:** Manufactures industrial AR solutions like the **ThinkReality A3** smart glasses.
- **Sony:** Developing AR glasses and currently leading in console-based mixed reality with **PSVR2**.
- **RealWear:** Specializes in voice-controlled, hands-free smart glasses (e.g., Navigator 520) for frontline industrial workers.
- **TCL RayNeo:** Manufactures consumer AR glasses, such as the RayNeo X3.
- **Viture:** Offers XR glasses geared toward gaming and media consumption.
- **Rokid:** Produces AR glasses for both enterprise visualization and consumer AI applications.

Specialized & Emerging Players

- **Brilliant Labs:** Known for introducing AI-powered, lightweight open-source smart glasses (Frame).
- **Solos:** Focuses on smart glasses with voice-based AI assistant features.
- **DigiLens:** Develops waveguide optics and smart glasses (EyeHUD) for wearable AR.
- **ThirdEye Gen:** Produces AR glasses for industrial and professional uses.
- **Merge:** Known for educational AR, such as the Merge Cube.

Key Industry Collaborations

- **Samsung:** Actively working on AR glasses, with plans to collaborate on AR/VR hardware.
- **Snap Inc. (Snapchat):** Produces Spectacles, which have evolved into standalone AR glasses.

These companies are heavily involved in the enterprise sector, with many focusing on "hand-free" applications, remote assistance, and 3D visualization to boost worker productivity, while others target the consumer market for gaming and entertainment.

To this we can also add a breakthrough in the development of displays of increasingly smaller sizes, but with the ability to show clarity as on a large screen. Physicists at the [University of Würzburg](#) have developed extremely small pixels using optical antennas and a clever design. Now, theoretically, augmented reality can even be placed in eye lenses.

The consumer we should be targeting next year is no longer the consumer of a standalone device—a mouse and keyboard. It will now be the consumer who requires a complete, ready-to-use solution, complete with an augmented reality display.

Even if we very modestly estimate this market at one million units per year, it will still prove to be a serious challenge requiring solutions in the area of expansion and reorganization of production.

Company valuation

Obviously, this process will require finding additional financing. This can be accomplished by holding the next round of fundraising. This will require first valuing the company and, based on this valuation, issuing shares and then offering them for sale. To achieve a good valuation, the company's performance indicators must improve.

The company's valuation will be based on the production and sales levels achieved by that point. Sales will now need to focus on the next consumer segment.

1. Digital Nomads and Remote Workers

This is the most obvious and ideal target audience. The device's primary benefit is its ability to provide a full-featured workspace without the need for a bulky laptop or a traditional keyboard.

- **Why it's a good fit:** These individuals are always on the move, working from cafes, co-working spaces, or public transport. The device's compact size, combined with the augmented reality interface, allows them to be productive anywhere. It solves the common problem of needing to carry a heavy setup while still requiring full text and mouse input.

2. Ergonomics Enthusiasts

The device is a prime example of an ergonomic solution, addressing the physical strain of repetitive motion.

- **Why it's a good fit:** People who are already interested in products that prevent or alleviate conditions like carpal tunnel syndrome would be highly receptive. The design encourages a natural hand position and reduces the need to constantly move between a mouse and a keyboard, which are key selling points for this market segment.

3. Early Adopters and Tech Enthusiasts

These are the individuals who are always on the lookout for the next big thing.

- **Why it's a good fit:** The device's combination of a unique physical design and an augmented reality interface makes it a conversation starter. This group would be excited to test and review a product that represents a new paradigm in human-computer interaction.

4. Specialized Professionals

Certain jobs require a high degree of precision and portability.

- **Why it's a good fit:** The device could be useful for professionals like graphic designers, architects, or video editors who need to zoom and scroll with precision. The dual scroll wheels and joysticks offer a level of control that might not be available on a standard mouse.

The next necessary step to improve the assessment will be the reorganization of production.

Reorganization of production

What should we strive for first?

The first thing we need to do is move away from manual assembly. This is a turning point where we need to stop solving production expansion problems by hiring personnel and instead restructure the entire technology toward automation. It's best to begin this process gradually and early. Without shutting down manual assembly in the first quarter of next year, we need to simultaneously begin automated assembly units for the most routine operations.

Assembly operations kind of 2 and 3 should be automated.

This should reduce the total time spent by 20-30%.

Ultimately, the goal of the reorganization will be to transition from producing the entire device to just its most basic components, outsourcing all final assembly to third-party companies. This should immediately increase production by an order of magnitude and reduce costs.

Therefore, it's worth limiting myself to just the keyboard sensor assembly, as everything else can be handled by any company that makes regular computer mice or similar devices.

I plan to take the first step in this direction in the first year. Assembling the side panels with batteries is fairly routine and could be outsourced to a company that previously manufactured, say, flashlights. However, the difference between the second year and the first should be that the entire sensor assembly will gradually be transitioned to automated assembly machines.

New sales price

The selling price of the device itself will now be determined according to a different principle, since the entire equipment set will be sold.

AR glasses in the US popular consumer models from [XREAL](#) and others often landing between **\$400 and \$600** for display-focused devices. Higher-end, standalone AR headsets with advanced spatial mapping typically cost between \$1,000 and \$2,500.

Key Price Breakdown (2025-2026):

- **Entry-Level/AI Glasses:** \$150 – \$700 (Focus on audio, camera, or basic notifications).
- **Mid-Range (Consumer AR):** \$300 – \$800 (Common for entertainment/display, e.g., Rokid, XREAL).
- **High-End/Prosumer:** \$500 – \$1,500+ (Standalone, advanced, or developer-focused).
- **Enterprise/Premium:** \$1,000 – \$3,000+ (Apple Vision Pro, specialized industrial gear).

Additional Costs:

- **Prescription Lenses:** Often require custom inserts costing \$100–\$400+.
- **Accessories:** Specialized cases, chargers, or specialized frames.
- **Subscriptions:** Some AR features may require ongoing fees.

Therefore, assuming that this mouse is fully capable of replacing all the complex additional functionality of augmented reality glasses, most devices can be downgraded to their basic configuration. The cost of the additional functionality, which will no longer be needed, significantly exceeds the price of the mouse. This is because all this functionality is complex and ineffective. Various devices that receive information from finger movements, head movements, and even eye blinks will never replace text. Because information obtained through gesture recognition, although also a language, is a very poor and primitive language, lagging behind spoken language as animal communication lags behind human communication. I'm not even talking about writing.

What will this give to the market for such devices?

Between 6 and 14 million different AR glasses were released or planned for release in 2025. Forecasts indicate this trend is set to grow steadily.

In comparison, mobile phones were produced in 1990, with approximately 11 million handsets, and 16 million in 1991, the year GSM was introduced. By 2000, there were already 420 million, despite the fact that a decline was predicted each subsequent year, but instead, steady growth continued throughout the 2000s. Predictions of decline were so persistent that companies like Ericsson, the industry leader at the time, exited the business, fearing a downturn and possible losses. They later tried to return, but Nokia had already firmly established their position.

In the 1990s and early 2000s, the advantages of cellular communications weren't so obvious. Making a call in the street wasn't particularly difficult through a wide network of public payphones, and many homes already had cordless phones. Cellular communications were expensive, and the phones themselves were primitive. The cellular network itself and coverage were also vastly underdeveloped. This encouraged cautious forecasts.

However, by 1992, 23 million cell phones had already been sold worldwide.

In 1993, 34M units were already sold.

I think it's quite reasonable to imagine, by analogy, that the number of AR glasses in 2027-2028 will be between 15 and 25 million units. If we take the price of a single set from \$700 to \$1,500, then the total market turnover will be between \$10.5 and \$37.5B. Furthermore, using the analogy with cell phones, we can assume that only growth is expected from here on out.

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